

1 Student Rollout (from Current Student)

Problem x

Student rollout from current policy

$$y_{1:T} \sim p_{\theta}(\cdot | x)$$

e.g., Solve for x
in $2x + 3 = 11$.

y_1

y_2

...

y_t

...

y_T

Evaluate the SAME frozen initial policy in two contexts



Unprivileged (no answer)
reference distribution

$q_t^0(\cdot)$



Privileged (with verified solution)
distribution

$q_t^+(\cdot)$



Outputs at each position t : $p_{\theta,t}(\cdot), q_t^0(\cdot), q_t^+(\cdot), y_t$

2 Sign-Conflict Candidate Gate

Token-level advantages (vs. student log-prob)

$$a_t^v = \log q_t^v(y_t) - \log p_{\theta,t}(y_t), v \in \{0, +\}$$

A position t is a candidate if all hold:

- ✓ **High entropy (student):** $H(p_{\theta,t}) \geq \tau_{\eta}$
- ✓ **Privileged suppresses:** $a_t^+ \leq -m$ (strongly disfavored)
- ✓ **Unprivileged does not suppress:** $a_t^0 \geq -m$ (not suppressed)

Entropy threshold (student):

$$\tau_{\eta} = Q_{\eta}(\{H(p_{\theta,j})\}_{j=1}^T) \quad (\text{e.g., } \eta = 0.75)$$

1

2

3

...

$t-1$

t

$t+1$

...

T

☒ Candidate (sign-conflict)

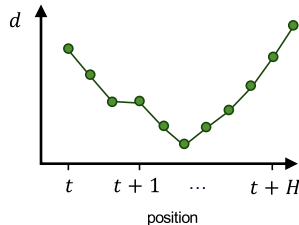
☐ Not candidate

3 Future-Recovery Score (Window Best Recovery)

Measure student-privileged-view divergence along the realized rollout

Divergence to privileged view

$$d_t = \text{JSD}(p_{\theta,t} \parallel q_t^+)$$



**Future-recovery score
(window best)**

$$\rho_t = d_t - \min_{1 \leq k \leq H} d_{t+k}$$

Large ρ_t : strong recovery
(small later divergence)

Small or negative ρ_t :
persistent / worsening

ρ_t can be negative if divergence increases at all future steps within the window.

4 Per-Trajectory Exact-Rank Calibration

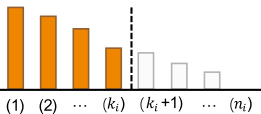
For trajectory i with n_i candidates and quantile $q \in [0, 1]$

Number routed to q^0 (unprivileged target):

$$k_i = \min\{n_i, \max\{1, \lceil (1 - q)n_i \rceil\}\}, \quad n_i > 0$$

Sort candidates by recovery score (high $\rightarrow$ low)

$$\rho(1) \geq \rho(2) \geq \dots \geq \rho(n_i)$$



Exact budget, zero tie excess

- ✓ Exactly k_i routed to q^0
- ✓ Exactly $n_i - k_i$ routed to q^+
- ✓ Zero per-trajectory budget error
- ✓ Zero tie excess

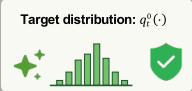
Tie-break by position and fixed hash for deterministic selection.
(Ensures reproducible ordering when ρ scores tie)

5 Context-Conditioned Target Routing

Targets for knowledge-distillation loss at position t

Top- k_i (highest recovery)
route to q^0 (unprivileged)

$$t \in S_t^0$$



Preserve exploration and
avoid answer-conditioned override

Remaining $n_i - k_i$
route to q^+ (privileged)

$$t \in S_t^+$$



Use solution context when
recovery is unlikely

Each trajectory receives exactly k_i unprivileged
and $n_i - k_i$ privileged targets.

6 ReGap: Three Counterfactual Quantities

Evaluation endpoints: Branch NLL / Recovery JSD / Verifier success

(a) Privileged-Context Advantage
(same (s, a) , different contexts)

fixed state-action pair (s, a)

Contexts:

Full

Rationale-masked

Answer-only

Shuffled rationale

No-reference (vanilla)

Quantifies benefit of privileged
context holding (s, a) fixed.

(b) Action Advantage
(same prefix, different actions)

fixed prefix u

actions at next position

Recovery-positive
action (exact-positive)

Matched-neutral action
(same format/length)

Random valid action

Quantifies action value
relative to matched controls.
(Not assumed to be correct.)

(c) Transferable Skill
(Held-out common-before DiD)

Held-out states (same action y_t)

Pre Post
(common) (held-out)

Positive (P) $\rightarrow$

Neutral (N) $\rightarrow$

Vanilla (V) $\rightarrow$

Common-before DiD

Δ_{P-N} Δ_{P-V}

Quantifies transferable recovery
skill beyond context and action.